

GILLIES ET AL. (2016)

Radiomics: Extracting more information from medical images using advanced feature analysis

European Journal of Cancer

□ Citations: 4,500+

□ Impact Factor: 7.275

□ Category: Foundational

ABSTRACT SUMMARY

□ Key Contribution: First comprehensive radiomics definition

• Introduced the term 'radiomics' and established its definition

• Proposed comprehensive radiomics workflow

• Identified four main feature categories

• Demonstrated clinical relevance in oncology

• Established reproducibility standards

METHODOLOGY ANALYSIS

□ RADIOMICS WORKFLOW

1. Image Acquisition: Standardized imaging protocols

2. Image Segmentation: Manual or automated ROI definition

3. Feature Extraction: Quantitative feature calculation

4. Feature Selection: Statistical and machine learning methods

□ FEATURE CATEGORIES

5. Model Building: Predictive model development

6. Feature Categories: Volume, surface area, compactness

• First-Order Features: Mean, variance, skewness, kurtosis

• Texture Features: GLCM, GLRLM, GLSZM matrices

• Higher-Order Features: Wavelet, fractal, filter-based

KEY CONTRIBUTIONS & IMPACT

□ MAJOR CONTRIBUTIONS

- First comprehensive definition of radiomics
- Established standardized radiomics workflow
- Introduced feature categorization system
- Demonstrated clinical relevance in multiple cancer types

□ CLINICAL IMPACT

- Established reproducibility and validation standards
- Enabled non-invasive tumor characterization development
- Facilitated personalized treatment planning
- Improved prognostic assessment accuracy
- Reduced need for invasive procedures
- Standardized imaging biomarker development

IMPLEMENTATION & RESULTS

□ □ IMPLEMENTATION DETAILS

- Dataset: Multi-center cancer imaging data
- Modalities: CT, MRI, PET imaging
- Features: 1000+ radiomics features extracted
- Validation: Cross-validation and independent testing

□ RESULTS SUMMARY

- Software: PyRadiomics library development
- Successfully extracted 1000+ radiomics features (validation Initiative)
- Demonstrated feature reproducibility across centers
- Established clinical relevance in multiple cancer types
- Achieved significant prognostic value
- Created standardized feature extraction protocols

FUTURE DIRECTIONS & LEGACY

□ □ LEGACY

- Foundation for entire radiomics field
- PyRadiomics library (most widely used radiomics software)
- IBSI standardization initiative
- Thousands of follow-up studies

□ FUTURE DIRECTIONS

- Clinical implementation in multiple centers
- Deep learning integration with radiomics
- Multi-modal radiomics analysis
- Real-time clinical implementation
- Regulatory approval for clinical use
- Standardization across imaging platforms